AUTHORISED ATTACK-SURFACE MONITORING

THREAT LENS

Authorised Attack-Surface Inventory and Continuous Change Monitor for Small Security Teams

TEAM CODE
A3-PS011-TC255

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PROJECT CONTRIBUTORS

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EXECUTIVE SUMMARY

Abstract & Problem Description

Abstract

Threat Lens is an authorised attack-surface monitoring platform designed to record approved assets, establish baseline snapshots, and detect state changes across scans.

Core Outcome:
Timely alert visibility with timestamped scan evidence.

Problem Description

- **Dynamic Exposure:** Continuous additions of subdomains, IPs, ports, and web endpoints expand attack surfaces.
- **Stale Manual Audits:** Periodic checks fail to capture exact timelines when unknown changes occur.
- **Raw Data Noise:** Asset lists fail to distinguish routine changes from potential security risks.

Delay in detecting exposure increases vulnerability windows.

Objective & Scope

Build a scope-aware prototype that inventories authorized assets, compares snapshots, and prioritizes notable changes.

SCOPE PRINCIPLE
Discovery is strictly limited to operator-authorized targets.

Background & Core Concepts

Attack Surface Definition

The full set of reachable entry points exposed by an organization's internal or external infrastructure.

- Subdomains
- IP Ranges
- Open Ports
- Active Services
- Web Endpoints

Expanded surfaces increase unauthorized entry risk.

Why Continuous Monitoring

- **Baseline Reference:** Establish ground truth for future comparison.
- **Delta Visibility:** Instantly highlight additions, removals, and drift.
- **Risk Context:** Prioritize changes that elevate security exposure.
- **Audit Proof:** Timestamped records simplify compliance validation.

Reduces Mean-Time-To-Detect (MTTD).

Operating Boundaries

Operates via scheduled or trigger-based scans within strictly approved scope parameters.

Decision Support Tool
Aids human operator analysis without autonomous exploitation.

Tested on synthetic & team-controlled targets.

STANDARDS & GUIDANCE

Literature & RFC Review

NIST SP 800-115

Testing Guide

Establishes planning, authorization, structured testing, and evidence reporting guidelines for controlled security assessments.

OWASP WSTG

Web Security

Provides structured methodology for assessing web endpoints; directly informs categorization of web-facing asset observations.

RFC 9116

security.txt

Defines standardized web formats for publishing security contact metadata; provides optional context during asset discovery.

CISA KEV Catalog

Intel Enrichment

Enables future threat enrichment mapping discovered services to known exploited vulnerabilities (post-MVP extension).

KEY LITERATURE TAKEAWAY

These standards guide safe assessment practices, asset organization, and evidence structure. Threat Lens applies these principles in a prototype setting without claiming formal compliance certification.

END-TO-END ARCHITECTURE

Proposed Solution & Core Modules

1. Scope Manager

Records approved domains, IP ranges, and scan boundaries.

2. Asset Discovery

Gathers permitted DNS, port, service, and web observations.

3. Inventory Baseline

Stores assets and timestamped snapshots for comparative analysis.

4. Change Engine

Detects new, removed, and modified services across snapshots.

5. Risk & Evidence

Assigns priority labels and attaches raw scan proof for alerts.

6. Dashboard & Reports

Presents change timeline, filters, metrics, and exportable reports.

Operational Example:
A baseline server exposes ports **80** and **443**. A later scan detects newly opened port **8080**. Threat Lens flags port 8080 as a high-priority change, attaches raw scan evidence, and alerts the operator for review.

VALUE OPTIMIZATION

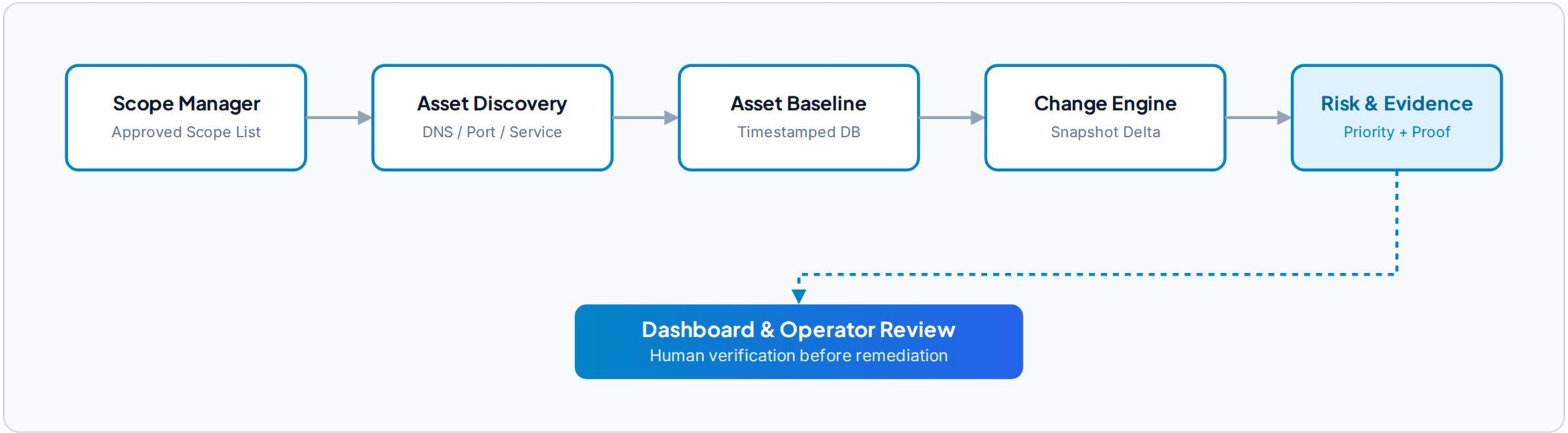
Solution Optimization & Comparison

Aspect	Manual / Basic Approach	Threat Lens Optimization
Asset Inventory	Ad-hoc lists or separate manual notes	**Centralized, timestamped snapshot database**
Change Detection	Manual comparison of raw scan diffs	**Automated delta analysis engine**
Prioritization	All observations treated with equal weight	**Contextual risk scoring & priority labels**
Evidence & Audit	Disconnected screenshots or log files	**Alerts directly linked to raw scan proof**
Scope Control	Informal tracking with high mistake risk	**Enforced allowlist with out-of-scope blocking**

Evaluation Measures: Discovery coverage on test scopes, change detection accuracy, scan execution time, evidence completeness, and zero out-of-scope scan breaches.

HIGH-LEVEL ARCHITECTURE

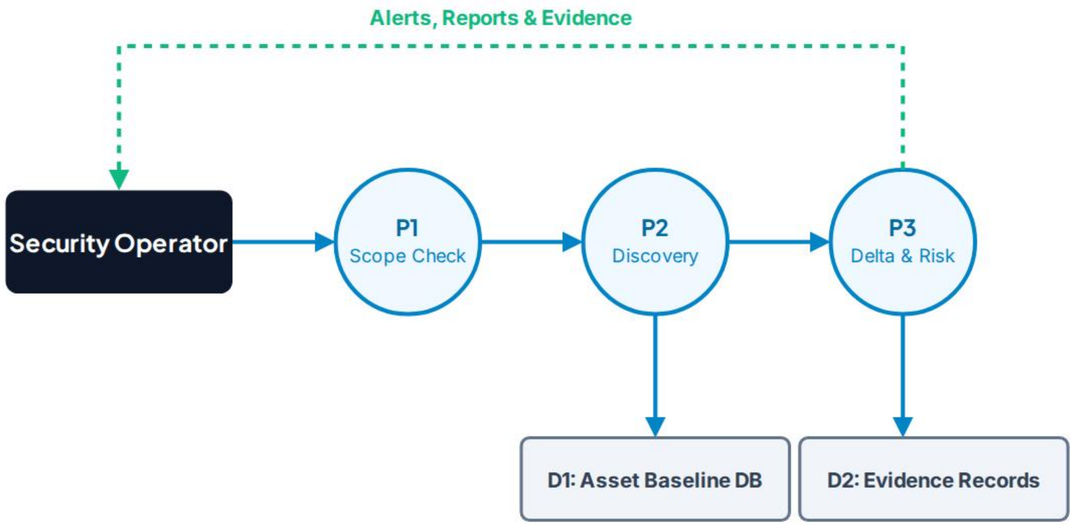
Solution Architecture & Workflow



The scope manager validates target authorization prior to scan execution. Discovered assets are saved as baseline snapshots; the change engine computes deltas, attaches evidence, and forwards priority alerts to the operator dashboard.

INFORMATION FLOW

Data Flow Diagram (DFD)



Primary Data Objects

1. Scope Definition:
Explicit list of approved target domains, IP ranges, and scan constraints.

2. Asset Snapshot:
Observed host state, open ports, web endpoints, and precise timestamp.

3. Delta Change Record:
Diff comparing previous vs. current state with assigned change category.

4. Evidence Trace:
Raw response output and timestamp linked directly to the alert record.

IMPLEMENTATION ROADMAP

Deployment Timeline & Phases

1
Requirements
Confirm PS-011 scope & test targets.

2
Design
Define modules, data model & DFD.

3
MVP Engine
Build scope check & discovery core.

4
Evidence & UI
Dashboard, alerts & report export.

5
Testing
Validate delta accuracy & scope rules.

6
Demo
Controlled lab demo & docs.

Agile Development Methodology:
Build a small, testable MVP first; validate core modules against controlled test targets; conduct progress reviews with the faculty mentor; then refine UI, proof generation, and final documentation.

❞ REFERENCES

Bibliography

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[3] **Foudil, E. and D. Foudil.** *RFC 9116: A File Format to Aid in Security.txt*, RFC Editor, 2022.
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[4] **Cybersecurity and Infrastructure Security Agency (CISA).** *Known Exploited Vulnerabilities (KEV) Catalog*.
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Note: These references provide structural guidance for security assessments and data handling. They do not imply formal endorsement or certification of the prototype.

THANK YOU

Threat Lens

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"Discover responsibly. Track changes. Keep evidence."